INDIVIDUAL WEEKLY PROJECT REPORTS

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Role: Documentation and Presentation
Project Title: Flight Delay Analysis and Visualization

WEEK 7 – Statistical Analysis and Results Documentation

During the seventh week, my main responsibility was documenting the statistical analysis performed as part of the flight delay project. The team focused on understanding relationships between numerical variables and examining the results obtained through correlation analysis and Principal Component Analysis (PCA).

I studied the purpose of correlation analysis and understood how it could be used to examine the relationship between departure delays and arrival delays. One of the important findings shared during the analysis was a strong positive correlation of approximately 0.971 between departure delay and arrival delay. I recorded this result and prepared an explanation of what it indicated about the relationship between the two variables.

I also focused on documenting PCA, which was included to explore the structure of the numerical data and identify the principal components that explain variation within the dataset. I worked on understanding the basic purpose of dimensionality reduction so that I could explain the technique clearly in the project report.

I coordinated with the team members responsible for statistical analysis to obtain the relevant observations and ensure that the documentation reflected the actual results. I organized the information into separate sections covering the purpose of each technique, its application, and the observations obtained.

In addition, I began preparing the results section of the report. I focused on presenting the numerical findings along with explanations so that readers could understand their significance without needing to interpret the calculations independently.

Outcome and Learning:

This week helped me improve my understanding of statistical analysis and technical report writing. I learned how to document statistical findings accurately and explain their relevance to the project's objectives.